Singapore School Kinderland

Secondary-1 (Sec 1) • Computer Science / Information Technology

TEACHER ANSWER KEY & STEP-BY-STEP WORKED SOLUTIONS (20 QUESTIONS)

Student Name: OFFICIAL TEACHER MASTER KEY	**Date:** _____
Topic: Text, Sound & Image File Size Calculations	**Total Questions: 20 (Total Marks: 68)**

PART 1: COMPLETE FORMULAS, CONVERSION UNITS & EXAM SHORTCUT TRICKS

1. Storage Capacity Units (Bits, Bytes, KiB/KB, MiB/MB) & Frequency Units (Hz, kHz, MHz)

Binary Units (IEC - Factor 1,024)	Decimal Units (SI - Factor 1,000)	Sound & Time Units (Hz, kHz, MHz)
• **4 bits** = 1 Nibble • **8 bits** = 1 Byte (B) • **1,024 Bytes** = 1 KiB (Kibibyte) • **1,024 KiB** = 1 MiB (Mebibyte) • **1,024 MiB** = 1 GiB (Gibibyte)	• **8 bits** = 1 Byte (B) • **1,000 Bytes** = 1 KB (Kilobyte) • **1,000 KB** = 1 MB (Megabyte) • **1,000,000 Bytes** = 1 MB • **1,000 MB** = 1 GB (Gigabyte)	• **1 Hz (Hertz)** = 1 sample per second • **1 kHz (Kilohertz)** = 1,000 Hz • **1 MHz (Megahertz)** = 1,000,000 Hz • **1 Minute** = 60 Seconds • **Mono** = 1 Channel | **Stereo** = 2 Channels

💡 **GOLDEN TRICK: To convert SMALL unit to BIG unit (bits → Bytes → KB/KiB → MB/MiB), always DIVIDE (÷). To convert BIG unit to SMALL unit, always MULTIPLY (×)!**

2. How to Calculate TEXT File Size

Text File Size (bits) = Number of Characters × Bits per Character | Size (Bytes) = Bits ÷ 8

- **Standard / Extended ASCII:** Uses **8 bits (1 Byte)** per character.
- **Unicode (16-bit):** Uses **16 bits (2 Bytes)** per character.
- **Exam Trap to Avoid:** Always count **spaces, numbers, and punctuation marks** as characters!

3. How to Calculate BITMAP IMAGE File Size

Image Size (bits) = Width (pixels) × Height (pixels) × Colour Depth (bits) | Size (Bytes) = Bits ÷ 8

Number of Colours	2 Colours (B&W)	4 Colours	8 Colours	16 Colours	256 Colours	65,536 Colours	16.7 Million (True Colour)
Colour Depth (bits)	**1 bit** (2^1)	**2 bits** (2^2)	**3 bits** (2^3)	**4 bits** (2^4)	**8 bits** (1 Byte)	**16 bits** (2 Bytes)	**24 bits** (3 Bytes)

4. How to Calculate DIGITAL SOUND File Size

Sound Size (bits) = Sample Rate (Hz) × Sample Resolution (bits) × Time (seconds) × Channels

- **Fast Division Trick:** Instead of multiplying by 8, 16, or 24 bits and dividing by 8 at the end, convert Sample Resolution into Bytes first: **8 bits = 1 Byte**, **16 bits = 2 Bytes**, **24 bits = 3 Bytes**!

SECTION A: UNIT CONVERSIONS (HZ, KHZ, MHZ, KB/KIB, MB/MIB)

Question 1 **[3 Marks]**

Convert the following frequency and sample rate values:
(a) 8 kHz into Hz
(b) 44.1 kHz into Hz
(c) 2.5 MHz into kHz and Hz

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Recall that 1 kHz = 1,000 Hz and 1 MHz = 1,000 kHz = 1,000,000 Hz.
- (a) 8 kHz = 8 × 1,000 = 8,000 Hz
- (b) 44.1 kHz = 44.1 × 1,000 = 44,100 Hz
- (c) 2.5 MHz = 2.5 × 1,000 = 2,500 kHz = 2,500,000 Hz

➔ Final Answer: (a) 8,000 Hz | (b) 44,100 Hz | (c) 2,500 kHz (or 2,500,000 Hz)

Question 2 **[3 Marks]**

Convert the following binary data measurements:
(a) 64 bits into Bytes
(b) 400 Bytes into bits
(c) 8,192 bits into Bytes, and then into KiB (Kibibytes)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Recall that 8 bits = 1 Byte and 1,024 Bytes = 1 KiB.
- (a) 64 bits ÷ 8 = 8 Bytes
- (b) 400 Bytes × 8 = 3,200 bits
- (c) 8,192 bits ÷ 8 = 1,024 Bytes --> 1,024 Bytes ÷ 1,024 = 1 KiB

➔ Final Answer: (a) 8 Bytes | (b) 3,200 bits | (c) 1,024 Bytes = 1 KiB

Question 3 **[3 Marks]**

Convert the following storage capacity units:
(a) 5 KiB into Bytes (using binary factor 1,024)
(b) 3 MB into Bytes (using decimal factor 1,000,000)
(c) 2,048 KiB into MiB

STEP-BY-STEP WORKING & SOLUTION:

- (a) 5 KiB × 1,024 = 5,120 Bytes
- (b) 3 MB × 1,000,000 = 3,000,000 Bytes
- (c) 2,048 KiB ÷ 1,024 = 2 MiB

➔ Final Answer: (a) 5,120 Bytes | (b) 3,000,000 Bytes | (c) 2 MiB

Question 4 **[2 Marks]**

Convert the following time durations and audio channel settings for sound calculations:
(a) 2.5 minutes into seconds
(b) State the number of channels for Mono audio and Stereo audio.

STEP-BY-STEP WORKING & SOLUTION:

- (a) Time in seconds = 2.5 × 60 = 150 seconds
- (b) Mono = 1 channel | Stereo = 2 channels

➔ Final Answer: (a) 150 seconds | (b) Mono = 1 channel, Stereo = 2 channels

SECTION B: TEXT FILE SIZE CALCULATIONS (ASCII & UNICODE)

Question 5 **[3 Marks]**

A student types the following phrase in a plain text editor using 8-bit ASCII encoding:
"HELLO SECONDARY 1"
Count the total number of characters (including spaces) and calculate the file size in bits and Bytes.

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Count characters in "HELLO SECONDARY 1":
- H-E-L-L-O (5) + [space] (1) + S-E-C-O-N-D-A-R-Y (9) + [space] (1) + 1 (1) = 17 characters.
- Step 2: In 8-bit ASCII, each character = 8 bits = 1 Byte.
- Size in bits = 17 × 8 = 136 bits
- Size in Bytes = 17 × 1 = 17 Bytes

➔ **Final Answer: 17 characters | 136 bits | 17 Bytes**

Question 6 **[3 Marks]**

A plain text essay contains 4,096 characters (including spaces and punctuation). It is saved using standard 8-bit ASCII encoding.
Calculate the size of the text file in:
(a) Bytes
(b) KiB (Kibibytes)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: 8-bit ASCII uses 1 Byte per character.
- (a) Size in Bytes = 4,096 × 1 = 4,096 Bytes
- (b) Size in KiB = 4,096 ÷ 1,024 = 4 KiB

➔ **Final Answer: (a) 4,096 Bytes | (b) 4 KiB**

Question 7 **[3 Marks]**

A multi-language webpage contains 3,500 characters and is saved using 16-bit Unicode encoding.
Calculate the file size in:
(a) bits
(b) Bytes
(c) KB (Kilobytes, dividing by 1,000)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: 16-bit Unicode = 16 bits (or 2 Bytes) per character.
- (a) Size in bits = 3,500 × 16 = 56,000 bits
- (b) Size in Bytes = 56,000 ÷ 8 (or 3,500 × 2) = 7,000 Bytes
- (c) Size in KB = 7,000 ÷ 1,000 = 7 KB

➔ **Final Answer: (a) 56,000 bits | (b) 7,000 Bytes | (c) 7 KB**

Question 8 **[4 Marks]**

A digital e-book has 10 pages. Each page has 32 lines, and each line has 64 characters (including spaces). The e-book uses 8-bit ASCII.
Calculate the total file size of the 10-page e-book in Bytes and in KiB.

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Total characters = 10 pages × 32 lines × 64 characters = 20,480 characters.
- Step 2: Since 8-bit ASCII = 1 Byte per character:
- Size in Bytes = 20,480 × 1 = 20,480 Bytes
- Step 3: Convert Bytes to KiB = 20,480 ÷ 1,024 = 20 KiB

➔ **Final Answer: 20,480 Bytes | 20 KiB**

SECTION C: BITMAP IMAGE FILE SIZE CALCULATIONS

Question 9 **[3 Marks]**

A small game icon has a resolution of 100 × 80 pixels and uses a colour depth of 8 bits per pixel.
Calculate the uncompressed image file size in:
(a) bits
(b) Bytes
(c) KB (dividing by 1,000)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Total pixels = Width × Height = 100 × 80 = 8,000 pixels.
- (a) Size in bits = 8,000 × 8 bits = 64,000 bits
- (b) Size in Bytes = 64,000 ÷ 8 = 8,000 Bytes
- (c) Size in KB = 8,000 ÷ 1,000 = 8 KB

➔ Final Answer: (a) 64,000 bits | (b) 8,000 Bytes | (c) 8 KB

Question 10 **[3 Marks]**

A black-and-white barcode image (2 colours only) has dimensions of 400 pixels wide by 200 pixels high.
(a) State the colour depth in bits for a 2-colour image.
(b) Calculate the file size of the barcode image in Bytes and KB.

STEP-BY-STEP WORKING & SOLUTION:

- (a) Since 2^1 = 2 colours, Colour Depth = 1 bit per pixel.
- (b) Size in bits = 400 × 200 × 1 = 80,000 bits
- Size in Bytes = 80,000 ÷ 8 = 10,000 Bytes
- Size in KB = 10,000 ÷ 1,000 = 10 KB

➔ Final Answer: (a) 1 bit | (b) 10,000 Bytes = 10 KB

Question 11 **[4 Marks]**

A retro arcade graphic has a resolution of 512 × 256 pixels and displays a maximum of 16 colours.
(a) How many bits per pixel (colour depth) are required for 16 colours?
(b) Calculate the exact file size of the graphic in KiB (dividing by 1,024).

STEP-BY-STEP WORKING & SOLUTION:

- (a) Since 2^4 = 16 colours, Colour Depth = 4 bits per pixel.
- (b) Size in bits = 512 × 256 × 4 = 524,288 bits
- Size in Bytes = 524,288 ÷ 8 = 65,536 Bytes
- Size in KiB = 65,536 ÷ 1,024 = 64 KiB

➔ Final Answer: (a) 4 bits | (b) 64 KiB

Question 12 **[3 Marks]**

A high-quality digital photograph has a resolution of 1,000 × 800 pixels and uses 24-bit True Colour.
Calculate the file size of the photograph in:
(a) Bytes
(b) MB (Megabytes, dividing by 1,000,000)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Trick! 24 bits = 24 ÷ 8 = 3 Bytes per pixel.
- Step 2: Total pixels = 1,000 × 800 = 800,000 pixels.
- (a) Size in Bytes = 800,000 × 3 Bytes = 2,400,000 Bytes
- (b) Size in MB = 2,400,000 ÷ 1,000,000 = 2.4 MB

➔ Final Answer: (a) 2,400,000 Bytes | (b) 2.4 MB

Question 13 [3 Marks]

A web banner has dimensions of 1,024 × 512 pixels and uses a 256-colour palette.
(a) Determine the colour depth in bits and Bytes for 256 colours.
(b) Calculate the file size of the banner in KiB.

STEP-BY-STEP WORKING & SOLUTION:

- (a) Since 2^8 = 256 colours, Colour Depth = 8 bits = 1 Byte per pixel.
- (b) Size in Bytes = 1,024 × 512 × 1 Byte = 524,288 Bytes
- Size in KiB = 524,288 ÷ 1,024 = 512 KiB

➔ Final Answer: (a) 8 bits (1 Byte) | (b) 512 KiB

Question 14 [4 Marks]

A designer saves 5 uncompressed images on a flash drive. Each image has a resolution of 2,000 × 1,000 pixels and a 16-bit colour depth.
Calculate the total storage space required for all 5 images in MB (dividing by 1,000,000).

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: 16-bit colour depth = 16 ÷ 8 = 2 Bytes per pixel.
- Step 2: Size of 1 image = 2,000 × 1,000 × 2 Bytes = 4,000,000 Bytes (4 MB).
- Step 3: Total size of 5 images = 4,000,000 × 5 = 20,000,000 Bytes
- Step 4: Convert to MB = 20,000,000 ÷ 1,000,000 = 20 MB

➔ Final Answer: 20,000,000 Bytes = 20 MB

SECTION D: DIGITAL SOUND FILE SIZE CALCULATIONS

Question 15 [3 Marks]

A 10-second mono voice note is recorded with a sample rate of 8,000 Hz and a sample resolution of 8 bits.
Calculate the file size in:
(a) bits
(b) Bytes
(c) KB (dividing by 1,000)

STEP-BY-STEP WORKING & SOLUTION:

- Formula: Sample Rate × Sample Resolution × Time × Channels
- (a) Size in bits = 8,000 × 8 × 10 × 1 (mono) = 640,000 bits
- (b) Size in Bytes = 640,000 ÷ 8 = 80,000 Bytes
- (c) Size in KB = 80,000 ÷ 1,000 = 80 KB

➔ Final Answer: (a) 640,000 bits | (b) 80,000 Bytes | (c) 80 KB

Question 16 [4 Marks]

A 30-second stereo sound clip is recorded at a sample rate of 20 kHz with a sample resolution of 16 bits.
Calculate the file size of the audio clip in:
(a) Bytes
(b) MB (dividing by 1,000,000)

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Convert 20 kHz to Hz = 20 × 1,000 = 20,000 Hz.
- Step 2: 16 bits = 2 Bytes per sample; Stereo = 2 channels.
- (a) Size in Bytes = 20,000 × 2 Bytes × 30 seconds × 2 channels = 2,400,000 Bytes
- (b) Size in MB = 2,400,000 ÷ 1,000,000 = 2.4 MB

➔ Final Answer: (a) 2,400,000 Bytes | (b) 2.4 MB

Question 17 [4 Marks]

A 1-minute mono podcast is recorded at a sample rate of 10 kHz and a sample resolution of 16 bits.
Calculate the total file size in Bytes and in KB (dividing by 1,000).

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: Convert 1 minute = 60 seconds; 10 kHz = 10,000 Hz.
- Step 2: 16 bits = 2 Bytes per sample; Mono = 1 channel.
- Step 3: Size in Bytes = 10,000 × 2 Bytes × 60 seconds × 1 = 1,200,000 Bytes
- Step 4: Size in KB = 1,200,000 ÷ 1,000 = 1,200 KB

➔ Final Answer: 1,200,000 Bytes | 1,200 KB (or 1.2 MB)

Question 18 [4 Marks]

A 20-second stereo song is recorded at CD quality with a sample rate of 44.1 kHz and a sample resolution of 16 bits.
Calculate the uncompressed audio file size in Bytes and MB (dividing by 1,000,000).

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: 44.1 kHz = 44,100 Hz; 16 bits = 2 Bytes; Stereo = 2 channels.
- Step 2: Size in Bytes = 44,100 × 2 Bytes × 20 seconds × 2 channels = 3,528,000 Bytes
- Step 3: Size in MB = 3,528,000 ÷ 1,000,000 = 3.528 MB

➔ Final Answer: 3,528,000 Bytes | 3.528 MB

Question 19 [4 Marks]

A 2-minute stereo background music track is sampled at 16,000 Hz with an 8-bit resolution.
Calculate the file size in Bytes and in MB (dividing by 1,000,000).

STEP-BY-STEP WORKING & SOLUTION:

- Step 1: 2 minutes = 2 × 60 = 120 seconds; 8 bits = 1 Byte; Stereo = 2 channels.
- Step 2: Size in Bytes = 16,000 × 1 Byte × 120 seconds × 2 channels = 3,840,000 Bytes
- Step 3: Size in MB = 3,840,000 ÷ 1,000,000 = 3.84 MB

➔ Final Answer: 3,840,000 Bytes | 3.84 MB

SECTION E: MASTER MULTIMEDIA CHALLENGE

Question 20 [5 Marks]

A student creates a multimedia slide containing three elements:
1. Text: 2,000 characters encoded in 8-bit ASCII.
2. Image: One bitmap picture of 500 × 400 pixels with 8-bit colour depth.
3. Audio: One 5-second mono voice clip sampled at 10 kHz with 8-bit resolution.
Calculate the combined total file size of all three elements in KB (dividing by 1,000).

STEP-BY-STEP WORKING & SOLUTION:

- 1. Text Size = 2,000 × 1 Byte = 2,000 Bytes (2 KB)
- 2. Image Size = 500 × 400 × 1 Byte (8-bit) = 200,000 Bytes (200 KB)
- 3. Audio Size = 10,000 Hz × 1 Byte (8-bit) × 5 seconds × 1 (mono) = 50,000 Bytes (50 KB)
- Total Size in Bytes = 2,000 + 200,000 + 50,000 = 252,000 Bytes
- Total Size in KB = 252,000 ÷ 1,000 = 252 KB

➔ Final Answer: Text: 2 KB + Image: 200 KB + Audio: 50 KB = Total: 252 KB (252,000 Bytes)